

Maya Maciel-Seidman

maya.maciels-seidman@duke.edu | (516) 946-3191

ORCID iD: [0009-0006-1758-0752](https://orcid.org/0009-0006-1758-0752) | GitHub: github.com/mlmac-seid

Education

2030 **PhD in Earth and Climate Sciences, Duke University**

Advisor: Prof. Jonathan C. Ryan

2025 **Bachelor of Arts in Earth and Environmental Sciences, Vanderbilt University**

Honors: *cum laude*

Minors: Data Science; Environmental and Sustainability Studies

Research Appointments

Naval Research Enterprise Internship Program (NREIP)

Washington, D.C.

U.S. Naval Research Laboratory Remote Sensing Division

May 2024–December 2025

Advisor: Dr. Trina Merrick

- Developed CNN models to delineate, categorize, and map ice wedge polygons on the Alaska North Slope.
- Conducted fieldwork in July 2025 in Utqiagvik, AK to ground truth remote sensing data.
- Built machine learning models to predict changes in active layer thickness on the Alaska North Slope.

Undergraduate Researcher

Nashville, TN

Climate, Health, and Energy Equity Lab (CHEEL), Vanderbilt University

January 2023–May 2025

Advisors: Dr. Zdravka Tzankova, Dr. Carol C. Ziegler

- Developed quantitative methodology to calculate carbon offset value of energy efficient retrofits to low-income housing.

Undergraduate Research Assistant

Nashville, TN

Mixed Intelligence Development for Programming Lab, Vanderbilt University

January 2022–June 2022

Advisor: Dr. Yu Huang

- Contributed to study design to determine the effect of ambient environment on programmer performance.

Research Intern

Raleigh, NC

Game2Learn Lab, North Carolina State University

June 2021–August 2021

Advisor: Dr. Tiffany Barnes

- Created and tested computing-infused K-12 curriculum using UC Berkeley's Snap! language.

Awards

- 2026 **Best Student Oral Presentation** – International Symposium on Artificial Intelligence in Glaciology
- 2026 **National Science Foundation Graduate Research Fellowship Honorable Mention**
- 2025 **Vanderbilt University Outstanding Leadership in Earth and Environmental Sciences Award**
- 2021–2025 **Cornelius Vanderbilt Scholarship** – Vanderbilt University full tuition merit scholarship and funded summer immersive experience

Peer-Reviewed Publications

Maciel-Seidman, M. L., Merrick, T.L., Liang, R., Grossman, S.M., Richards, IV, D.F., Vermillion, M., Abelev, A. (2026) Ice-Wedge Polygon Geomorphometry: A Comparative Assessment Using Deep Learning and Repeat Lidar. *Scientific Reports* (under review)

Richards, IV, D.F., Merrick, T.L., Abelev, A., Vermillion, M., **Maciel-Seidman, M.L.** and Grossman, S.M. (2026), Remote Sensing-Based Framework for Detecting and Interpreting Permafrost Terrain Hydrologic Connectivity, *Frontiers in Earth Science*, doi.org/10.3389/feart.2026.1846337

Karas, Z., Page, D., **Maciel-Seidman, M.**, Sharafi, Z. and Huang, Y. (2026), The Influence of Environmental Odors on Student Programmers During Code Comprehension and Code Writing, *ACM Transactions on Software Engineering and Methodology*, doi.org/10.1145/3822406

Maciel-Seidman M., Tzankova Z., Ziegler C.C., Lele A., Lu S., Yan Y. and Muchira J.M. (2024), Mobilizing carbon offsetting to reduce energy cost burdens: a new approach for calculating and monetizing the offset value of energy efficiency upgrades to low-income housing, *Frontiers in Energy Research*, doi.org/10.3389/fenrg.2024.1437560

Maciel-Seidman, M. L., Channapatna, R. and Channapatna, S. (2020), Soil Moisture Sensing Robot: A Novel Agricultural Device, *Journal of Dawning Research*, <http://dawningresearch.org/Papers/2020/JDR2020002.pdf>

Conference Presentations

- 2026 **Maciel-Seidman, M.** and Ryan, J., Modeling Greenland Ice Sheet catchment-scale meltwater runoff using transfer learning, *International Symposium on Artificial Intelligence in Glaciology, Hanover, New Hampshire*
- 2026 Richards IV, D., Merrick, T., Liang, R., Abelev, A., Vermillion, M., **Maciel-Seidman, M.** and Grossman, S., Remote Sensing-Based Framework for Detecting and Interpreting Permafrost Terrain Hydrologic Connectivity, *EGU General Assembly, Vienna, Austria*
- 2024 **Maciel-Seidman, M. L.**, Merrick, T. L., Grossman, S. M., Richards, D. F., Abelev, A., Vermillion, M. S. and Liang, R. T., Analysis of Active Layer Thickness and Climate Data at Utqiagvik, Alaska with Random Forest and Multiple Linear Regression Algorithms, *AGU Annual Meeting, Washington, DC*

- 2024 Merrick, T. L., Richards, D. F., Grossman, S. M., **Maciel-Seidman, M. L.**, Abelev, A., Vermillion, M. S. and Liang, R. T., Arctic ice wedge polygon characterization using multilevel remote sensing, *AGU Annual Meeting, Washington, DC*
- 2024 Richards, D.F., Merrick, T. L., Grossman, S. M., **Maciel-Seidman, M. L.**, Abelev, A., Vermillion, M. S. and Liang, R. T., Remote Sensing of Periglacial Trough Structure Variability and Connections to Ice-Wedge Degradation, *AGU Annual Meeting, Washington, DC*
- 2024 Grossman, S. M., Merrick, T. L., Richards, D. F., **Maciel-Seidman, M. L.**, Abelev, A., Vermillion, M. S. and Liang, R. T., Multilevel remote sensing investigation of Tundra surface variability in Utqiagvik, Alaska, *AGU Annual Meeting, Washington, DC*

Software and Datasets

Maciel-Seidman, M., Merrick, T., Liang, R., Grossman, S., Richards, IV, D., Vermillion, M. and Abelev, A. (2026), Ice-Wedge Polygon Geomorphometry: A Comparative Assessment Using Deep Learning and Repeat Lidar Surveys, *Zenodo*, doi.org/10.5281/zenodo.19578055

Maciel-Seidman, M. (2026), Parallelized Greenland Ice Sheet Basin Data Query from MAR, *Zenodo*, doi.org/10.5281/zenodo.21263495

Fieldwork

- July 2025 **Utqiagvik, Alaska** – Carried out flight planning and UAS safety checks to enable collection of lidar data and hyperspectral and RGB imagery of ice wedge polygons. Measured soil active layer thickness, logged Trimble GeoXH points, collected vegetation and soil samples, and installed an array of soil moisture sensors and an automatic weather station. Conducted this fieldwork for a U.S. Naval Research Laboratory project to quantify and characterize spatiotemporal changes in Arctic periglacial terrain.
- Sept 2024 **Collegiate Peaks, Colorado** – Collected till samples, measured clast sizes, and calculated angle of repose to relatively date moraines from the Bull Lake and Pinedale (Pleistocene) glaciations. Conducted this fieldwork for Vanderbilt University course EES 4440: Glacial Geology.
- Aug–Dec 2023 **Tennessee State University Agricultural Research and Education Center** – Augered soil samples for hydrocarbon analysis and created soil profiles. Collected and filtered water samples from research wells to analyze oxygen and hydrogen isotopes, dissolved inorganic carbon, and volatile organic compounds. Conducted this fieldwork for Vanderbilt University course EES 3280: Environmental Geochemistry.

Teaching

Vanderbilt University, Nashville, TN

- Spring 2025 Grader, DS 1000, How Data Shape Our World
- Fall 2024 Grader, DS 1000, How Data Shape Our World
- Spring 2024 Grader, DS 1000, How Data Shape Our World
- Fall 2023 Teaching Assistant, ENVS 1101, Foundations of Climate Studies
- Fall 2023 Grader, DS 1000, How Data Shape Our World
- Spring 2023 Grader, DS 1000, How Data Shape Our World

Fall 2022 Grader, DS 1000, How Data Shape Our World

Professional Memberships and Affiliations

International Glaciological Society (2026–present)
Association of Polar Early Career Scientists (2025–present)
International Association of Cryospheric Sciences (2025–present)
American Geophysical Union (2024–present)
Phi Sigma Rho – Alpha Eta Chapter (2022–present)
National Center for Women & Information Technology (2020–present)

Relevant Courses

Glaciology: Glacial Geology; Field Investigations: Colorado
Computing: Programming and Problem Solving with Python; Program Design and Data Structures;
Data Science: How Data Shape Our World; Applied Machine Learning; Fundamentals of Data Science;
Agent-Based Modeling; Geospatial Data Science; Environmental Spatial Data Analysis; Software
Engineering for Spatial Data Analysis
Wilderness Medicine: National Outdoor Leadership School (NOLS) Wilderness First Aid

Skills

Computing: Python, R, Bash, MATLAB, C++, Java, NetLogo, GNU nano, Vim, Git, GitHub, High-
Performance Computing, Slurm, MPI
Geospatial Software: QGIS, ArcGIS Pro, GeoDa, Google Earth Pro, GeoMapApp, QGroundControl
Field equipment/techniques: Hach FH950 velocity flow meter, lightweight deflectometer, Trimble
GeoXH, ASD FieldSpec, tile probe, Kestrel 5500 Weather Meter, soil auger, clinometer
CAD and 3D Printing Software: Solid Edge, Autodesk Inventor, Ultimaker Cura, Bambu Studio